

Final Exam - DevOps Submission (Task 4)

Student Information

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Task 4: Jenkins CI/CD Pipeline Integration

I have successfully designed, configured, and integrated a complete Jenkins CI/CD pipeline using a declarative **Jenkinsfile** for the multi-container Spring Boot and MySQL application.

Below are the details, configurations, and verification steps demonstrating the successful implementation of this task.

1. Jenkins Pipeline Execution (Finished: SUCCESS)

The pipeline runs automatically on the Jenkins agent labeled **SpringBoot**. It successfully performs the following steps:

1. **Checkout:** Pulls the latest code from the GitHub repository branch **finalexam**.
2. **Build & Test:** Clean-builds the application using Gradle and executes all JUnit tests against the SQLite database.
3. **Deploy & Database Backup:**
 - Copies the newly built **demo-0.0.1-SNAPSHOT.jar** directly into the running **web** container.
 - Restarts the **web** container to apply the new build.
 - Performs a database dump of the MySQL database **A-PHE_Rithika-db** from the **db** container and saves it as **backup.sql** on the host machine.

 Jenkins Pipeline Success

2. Jenkinsfile Pipeline Configuration

The declarative pipeline script is configured to dynamically handle execution on the agent environment, performing native operations for build, deployment, and backup.

```
pipeline {
  agent { label 'SpringBoot' }

  triggers {
    pollSCM('*/5 * * * *')
  }
}
```

```

    stages {
        stage('Checkout') {
            steps {
                checkout scm
            }
        }

        stage('Build & Test') {
            steps {
                script {
                    if (isUnix()) {
                        sh 'chmod +x gradlew'
                        sh './gradlew clean build'
                    } else {
                        withEnv(["JAVA_HOME=C:\\Program Files\\Java\\jdk-25.0.3"])
                        bat 'gradlew.bat clean build'
                    }
                }
            }
        }

        stage('Deploy via Ansible') {
            steps {
                script {
                    if (isUnix()) {
                        sh 'ansible-playbook -i inventory.ini playbook.yml'
                    } else {
                        // Copy the newly built JAR directly into the running web
                        container
                        bat 'docker cp build\\libs\\demo-0.0.1-SNAPSHOT.jar
web:/app/build/libs/demo-0.0.1-SNAPSHOT.jar'
                        // Restart the container to apply the new JAR
                        bat 'docker restart web'
                        // Backup MySQL database to local file
                        bat 'docker exec db bash -c "mysqldump -u root -pHello@123
A-PHE_Rithika-db" > backup.sql'
                    }
                }
            }
        }
    }

    post {
        failure {
            // Send email alerts using Jenkins emailxt plugin on build or test
            failure
            emailxt (
                subject: "Build Failure in Jenkins: ${env.JOB_NAME} - Build
#${env.BUILD_NUMBER}",
                body: """"Hello,

The Jenkins build #${env.BUILD_NUMBER} for job '${env.JOB_NAME}' has FAILED.

```

Details:

- Build URL: \${env.BUILD_URL}
- Console Output: \${env.BUILD_URL}console
- Git Branch: \${env.GIT_BRANCH}

Please investigate and resolve the issue.

Regards,

Jenkins CI/CD System""",

to: 'srengty@gmail.com',

recipientProviders: [

[\$class: 'CulpritsRecipientProvider'],

[\$class: 'DevelopersRecipientProvider']

]

)

}

}

}

3. Pipeline Console Output Verification

As verified in the console log of Build #9:

- **Git checkout** successfully retrieved the code on branch **finalexam**.
- **Gradle clean build** executed successfully inside the agent environment in 22 seconds.
- **Deploy bat commands** ran successfully, copying the artifact and restarting the Spring Boot container.
- **MySQL dump** backed up the database data into **backup.sql** without issues.
- The pipeline status finished with a clean **Finished: SUCCESS**.

Submission Details

- **Repository URL:** <https://github.com/Rithika11-ui/I4A-FirstProjectJenkins.git>
- **Branch:** **finalexam**